7.4.1 Power supply input from ST-LINK/V2-1 USB connector

The STM32 Nucleo-144 board and shield can be powered from the ST-LINK USB connector CN1 (U5V), by placing a jumper between the pins 3 and 4 of JP3, as shown in *Table 8: Power related jumper*. Note that only the ST-LINK part is power supplied before the USB enumeration, as the host PC only provides 100 mA to the board at that time. During the USB enumeration, the STM32 Nucleo-144 board requires 300mA of current to the host PC. If the host is able to provide the required power, the targeted STM32 microcontroller is powered and the green LED LD6 is turned ON, thus the STM32 Nucleo-144 board and its shield consume a maximum of 300 mA current, not more. If the host is not able to provide the required current, the targeted STM32 microcontroller and the extension boards are not power supplied. As a consequence the green LED LD6 stays turned OFF. In such case it is mandatory to use an external power supply as explained in the next section.

After the USB enumeration succeeds, the ST-LINK U5V power is enabled, by asserting the PWR_EN pin. This pin is connected to a power switch (ST890), which powers the board. This power switch features also a current limitation to protect the PC in case of short-circuit on board. If an overcurrent (more than 500 mA) happens on board, the red LED LD5 is lit.

JP1 is configured according to the maximum current consumption of the board when powered by USB (U5V). JP1 jumper can be set to ON to inform the host PC that the maximum current consumption does not exceed 100 mA (including potential extension board or ST Zio shield). In such condition USB enumeration will always succeed, since no more than 100 mA is requested to the PC. Possible configurations of JP1 are summarized in *Table 6*.

Table 6. JP1 configuration table

Jumper state	Power supply	Allowed current
JP1 jumper OFF	USB power through CN1	300 mA max
JP1 jumper ON		100 mA max
JP1 jumper (do not care)	V_{IN}, +3.3 V, +5 V power	For current limitation refer to *Table 8*

Caution: In case the maximum current consumption of the STM32 Nucleo-144 board and its shield boards exceeds 300 mA, it is mandatory to power the STM32 Nucleo-144 board with an external power supply connected to E5V, V_{IN} or +3.3 V.

Note: In case the board is powered by a USB charger, there is no USB enumeration, so the green LED LD6 stays in OFF state permanently and the target STM32 is not powered. In this specific case the jumper JP1 must be set to ON, to allow the board to be powered anyway. But in any case the current is limited to 500 mA by U4 (ST890).

7.4.2 External power supply inputs

The Nucleo-144 board and its shield boards can be powered in three different ways from an external power supply, depending on the voltage used. The three power sources are summarized in the *Table 7*.

When STM32 Nucleo-144 board is power supplied by V_{IN} or E5V, the jumper configuration must be the following:

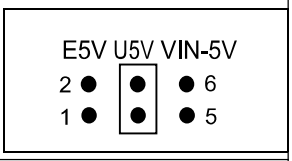
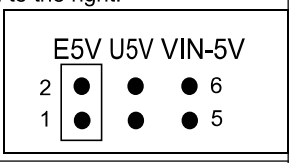
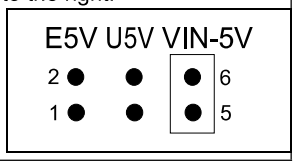
- Jumper JP3 on pin 1 and pin 2 for E5V or jumper JP3 on pin 5 and pin 6 for V_{IN}
- Jumper JP1 OFF

Table 7. External power sources

Input power name	Connector pins	Voltage range	Max current	Limitation
V_{IN}	CN8 pin 15 CN11 pin 24	7 V to 12 V	800 mA	From 7 V to 12 V only and input current capability is linked to input voltage: 800 mA input current when V_{IN}=7 V 450 mA input current when 7 V<V_{IN}<9 V 250 mA input current when 9 V<V_{IN}<12 V
E5V	CN11 pin 6	4.75 V to 5.25 V	500 mA	-
+3.3 V	CN8 pin 7 CN11 pin 16	3 V to 3.6 V	-	Two possibilities: ST-LINK PCB is cut SB3 and SB111 OFF (ST-LINK not powered)

The 5 V power source is selected by the jumper JP3 as shown in *Table 8*.

Table 8. Power related jumper

Jumper	Description
JP3	U5V (ST-LINK V_{BUS}) is used as power source when JP3 is set as shown to the right (Default setting) 
	E5V is used as power source when JP3 is set as shown to the right: 
	V_{IN} is used as power source when JP3 is set as shown to the right: 

Using V_{IN} or E5V as an external power supply

When powered by V_{IN} or E5V, it is still possible to use the ST-LINK for programming or debugging only, but it is mandatory to power the board first using V_{IN} or E5V, then to connect the USB cable to the PC. In this way the enumeration succeeds, thanks to the external power source.

The following power-sequence procedure must be respected:

1. Connect jumper JP3 between pin 1 and pin 2 for E5V or between pin 5 and pin 6 for V_{IN}
2. Check that JP1 is removed
3. Connect the external power source to V_{IN} or E5V
4. Power on the external power supply 7 V< V_{IN} < 12 V to V_{IN}, or 5 V for E5V
5. Check that the green LED LD6 is turned ON
6. Connect the PC to the USB connector CN1

If this order is not respected, the board may be powered by USB (U5V) first, then by V_{IN} or E5V as the following risks may be encountered:

1. If more than 300 mA current is needed by the board, the PC may be damaged or the current supplied can be limited by the PC. As a consequence the board is not powered correctly.
2. 300 mA is requested at enumeration (since JP1 must be OFF) so there is risk that the request is rejected and the enumeration does not succeed if the PC cannot provide such current. Consequently the board is not power supplied (LED LD6 remains OFF).

External power supply input: + 3.3 V

When the 3.3 V is provided by a shield board, it is interesting to use the +3.3 V (CN8 pin 7 or CN11 pin 16) directly as power input. In this case the programming and debugging features are not available, since the ST-LINK is not powered.

When the board is powered with +3.3 V, two different configurations are possible:

- ST-LINK is removed (PCB cut)
- SB3 (3.3 V regulator) and SB111 (NRST) are OFF.

7.4.3 External power supply output

When powered by USB, V_{IN} or E5V, the +5 V (CN8 pin 9 or CN11 pin 18) can be used as output power supply for an ST Zio shield or an extension board. In this case, the maximum current of the power source specified in *Table 7: External power sources* must be respected.

The +3.3 V (CN8 pin 7 or CN11 pin 16) can be used also as power supply output. The current is limited by the maximum current capability of the regulator U6 (500 mA max).

7.5 LEDs

User LD1: a green user LED is connected to the STM32 I/O PB0 (SB120 ON and SB119 OFF) or PA5 (SB119 ON and SB120 OFF) corresponding to the ST Zio D13.

User LD2: a blue user LED is connected to PB7.

User LD3: a red user LED is connected to PB14.

These user LEDs are on when the I/O is HIGH value, and are off when the I/O is LOW.

LD4 COM: the tricolor LED LD4 (green, orange, red) provides information about ST-LINK communication status. LD4 default color is red. LD4 turns to green to indicate that communication is in progress between the PC and the ST-LINK/V2-1, with the following setup:

- Slow blinking red/off: at power-on before USB initialization
- Fast blinking red/off: after the first correct communication between PC and ST-LINK/V2-1 (enumeration)
- Red LED on: when the initialization between the PC and ST-LINK/V2-1 is complete
- Green LED on: after a successful target communication initialization
- Blinking red/green: during communication with target
- Green on: communication finished and successful
- Orange on: communication failure

LD5 USB power fault: LD5 indicates that the board power consumption on USB exceeds 500 mA, consequently the user must power the board using an external power supply.

LD6 PWR: the green LED indicates that the STM32 part is powered and +5 V power is available on CN8 pin 9 and CN11 pin 18.

LD7 and LD8 USB FS: refer to *Section 7.10: USB OTG FS or device*.

7.6 Push-buttons

B1 USER: the user button is connected to the I/O PC13 by default (Tamper support, SB173 ON and SB180 OFF) or PA0 (Wakeup support, SB180 ON and SB173 OFF) of the STM32 microcontroller.

B2 RESET: this push-button is connected to NRST and is used to RESET the STM32 microcontroller.

7.7 JP5 (I_{DD})

Jumper JP5, labeled IDD, is used to measure the STM32 microcontroller consumption by removing the jumper and by connecting an ammeter:

- JP5 ON: STM32 is powered (default)
- JP5 OFF: an ammeter must be connected to measure the STM32 current. If there is no ammeter, the STM32 is not powered

To get a correct current consumption, the Ethernet PHY must be set in power-down mode or SB13 must be removed. Refer to *Section 7.11: Ethernet* for details.